

Table 3 - Comparative Summary of Compact Heat Exchanger Features

Type of Heat Exchanger	Area Density (m ² /m ³) (0)	Stream Types (1)	Materials (3)	Temperature Range (°C)	Maximum Pressure (bar) (2)	Fluid Limitations	Cleaning Methods	Corrosion Resistance (18)	Multi-stream Capability	Multi-pass Capability
Plate and frame (Gasketed)	→200	liquid-liquid gas-liquid two-phase	s/s, Ti, Incoloy Hastelloy, graphite, polymer	-25 to +175 Special -35 to +200	Normal 25 Special 40	Limited by gasket type. Not normal for gases	Mechanical (14) Chemical	Good (7)	Yes (9)	Yes
Partially welded plate	→200	gas-liquid liquid-liquid two-phase	s/s, Ti, Incoloy Hastelloy	-35 to +200	25	Few - Some types need clean fluids	Mechanical (4, 14) Chemical (6)	Good (7)	No	Yes
Fully welded plate (AlfaRex)	→200	gas-gas gas-liquid liquid-liquid two-phase	s/s, Ti, Ni alloys	-50 to +650	40	Few - Some types need clean fluid	Chemical	Excellent	No	Yes
Brazed plate	→200	liquid-liquid two-phase	s/s	Cu braze -195 to +220 Ni braze →400	Cu braze 30 Ni braze 16	Must be compatible with braze	Chemical (5)	Good (8)	No	No (10)
Bavex plate	200 - 300	gas-gas gas-liquid liquid-liquid two-phase	s/s, Ni, Cu, Ti, Special steels	-200 to +900	60	Few	Mechanical (11, 9) Chemical	Good	In principle	Yes
Platular plate	200	gas-gas gas-liquid liquid-liquid two-phase	s/s, Ti, Hastelloy, Ni alloys	-180 to +700	40	Few	Mechanical (12, 14)	Good	Yes (13)	Yes
Compabloc plate	→300	liquid-liquid two-phase	s/s, Ti, Incoloy	→300	32	Few	Mechanical (14)	Good	Not usually	Yes
Packinox Plate	→300	gas-gas gas-liquid liquid-liquid two-phase	s/s, Ti, Hastelloy, Inconel	-200 to +700	300	Few	Mechanical (16, 14)	Good	Yes (9)	Yes
Spiral	→200	gas-liquid liquid-liquid two-phase	c/s, s/s, Ti, Incoloy, Hastelloy	→400 Special →850	30	Few	Mechanical (14)	Good	No	No
Brazed plate-fin	800 - 1,500	gas-gas gas-liquid liquid-liquid two-phase	Al, s/s, Ti Ni alloy	Al -270 to +200 s/s cryogenic to +650	120	Low fouling Many limitations with Al	Chemical	Good	Yes	Yes

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Diffusion-bonded plate-fin	700 - 800	gas-gas gas-liquid liquid-liquid two-phase	Ti, s/s, Ni	→400	200	Low fouling	Chemical	Excellent	Yes	Yes
Printed circuit	200 - 5,000	gas-gas gas-liquid liquid-liquid two-phase	s/s, Ni, Ni alloys Ti	-200 to +900	500	Low fouling	Chemical	Excellent	Yes	Yes
Polymer - compact shell and tube	→275	liquid-liquid	Teflon	→200	11	Few	Water wash	Excellent	No	No
Plate and shell	→200	liquid-liquid	s/s, Ti, (shell also in c/s) (15)	-200 to +900	100	Few	Mechanical (16, 14) Chemical (17)	Good	No	Yes
Shell and tube (19)	→100	gas-gas gas-liquid liquid-liquid two-phase	s/s, Ti, (shell also in c/s), many different materials	-100 to +600	Shell 300 Tubes 1400	Few	Mechanical (16, 14) Chemical (17)	Good	No	Yes

Notes for Table 3

- = up to, s/s = stainless steel, Ti = titanium, Ni = nickel, Al = aluminium, Cu = copper, c/s = carbon steel
- (0) Area includes the secondary surface (such as fins)
- (2) The maximum pressure capability is unlikely to occur at the higher operating temperatures, and assumes no pressure/stress-related corrosion
- (4) On gasket side
- (6) On welded side
- (8) Function of braze as well as plate material
- (10) Not in a single unit
- (12) Only when flanged access provided, otherwise chemical cleaning
- (14) Can be dismantled
- (16) On shell side
- (18) Primarily a function of construction materials rather than the exchanger type
- (1) Two-phase includes boiling and condensing duties
- (3) Other special alloys are frequently available
- (5) Ensure compatibility with copper braze
- (7) Function of gasket as well as plate material
- (9) Not common
- (11) On tube side
- (13) Five fluids maximum
- (15) Shell may be composed of polymeric material
- (17) On plate or tube side
- (19) Not a compact exchanger technology (given for comparison)